Quantum Technologies

Jacq Romero

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Acknowledgement of Country

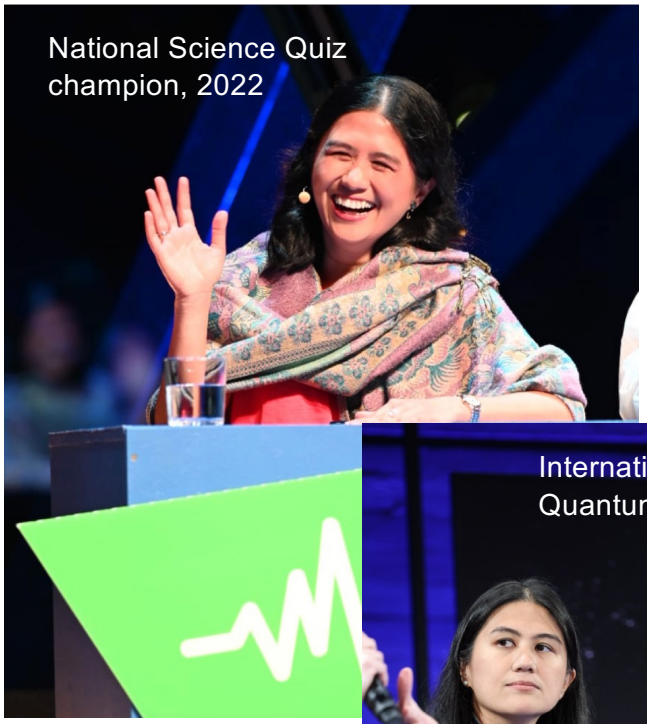
A Guidance Through Time
by Casey Coolwell and Kyra Mancktelow



The University of Queensland acknowledges the Traditional Owners and their custodianship of the lands on which we meet.

We pay our respects to their Ancestors and their descendants, who continue cultural and spiritual connections to Country.

We recognise their valuable contributions to Australian and global society.



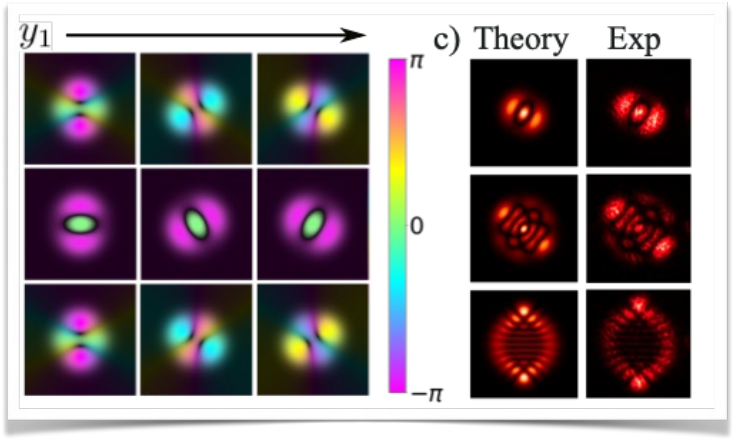


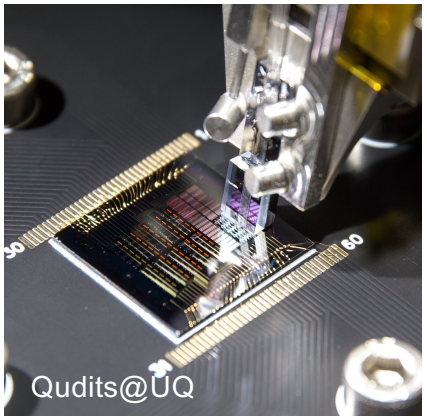
A/ Prof Jacq Romero (she/her)
Week 2, 3, 11
quantum information, communication, computation, photonic qubits

Born in the Phillippines

BSc.	University of the Philippines *magna cum laude*	2005
MSc	University of the Philippines.	2007
PhD	University of Glasgow	2013
Postdoc	University of Glasgow	2013–15
Postdoc	University of Queensland	2015–19

https://www.quantum-atbp.org





My research is in all things quantum using photons: quantum foundations, communication, and computation

Some housekeeping

Consultation time: Mondays 1 to 2 pm, or by appointment, 6-327

Use the Ed discussion board to discuss with your peers. Ed is monitored by the course staff but do not expect immediate response from them (24 to 48 hours is typical).

Jacq's "SAP": does not cope well in online chat environments, she prefers synchronous, 3D, in-person discussions.

The learning environment is everybody's responsibility—let us minimise distractions and be respectful and inclusive at all times.

Quantum technologies



Quantum communications leads to secure communications by exposing eavesdroppers.



Quantum sensors provide very high sensitivity and spatial resolution.



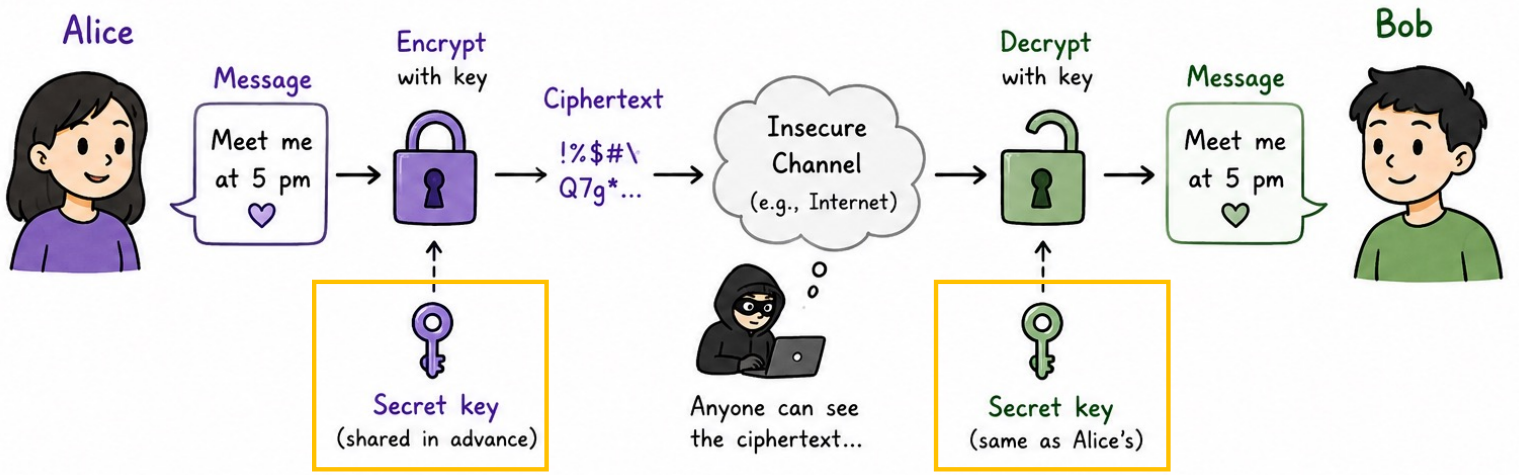
Quantum computing leads to a computational advantage for some specific problems.

Images created using Dall-E 3

What is Cryptography?

Cryptography is the science of keeping information secure.

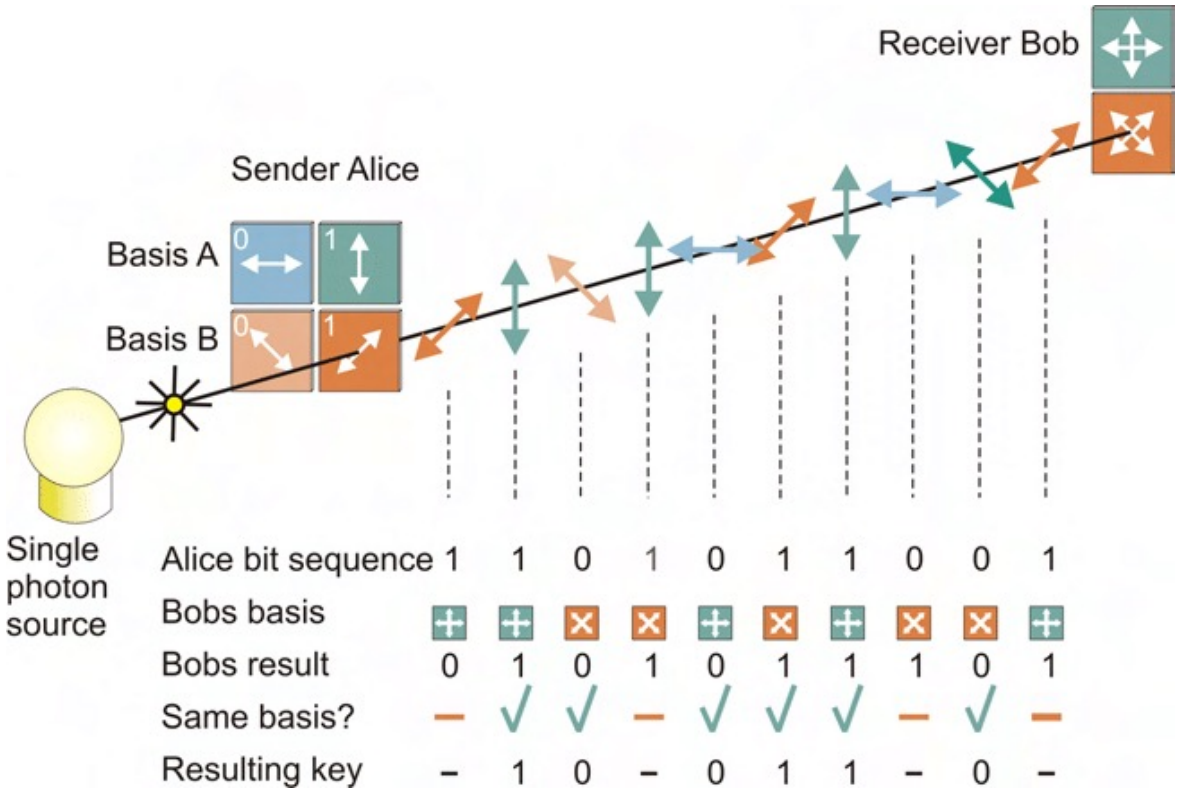
It allows people to communicate over insecure channels
so that only the intended recipient can understand the message.



Cryptography turns readable information into unreadable form
so that only the right person, with the right key, can read it.

Cartoon created using ChatGPT

BB84 protocol (Bennett and Brassard, 1984)



Eve's strategy: intercept Alice's photon, prepare a photon to send to Bob!

… but there is *no cloning* in quantum mechanics…

If there's Eve, Alice and Bob find out, because they will find errors even for cases when they both chose the same basis.

Eve's presence is found from statistical analysis of outcomes.

Stock

Quantum key distribution ~ 2017





Intercontinental quantum communication via Micius satellite, over 7,600 km, between China and Austria.

The most secure video call… ~kbps rate

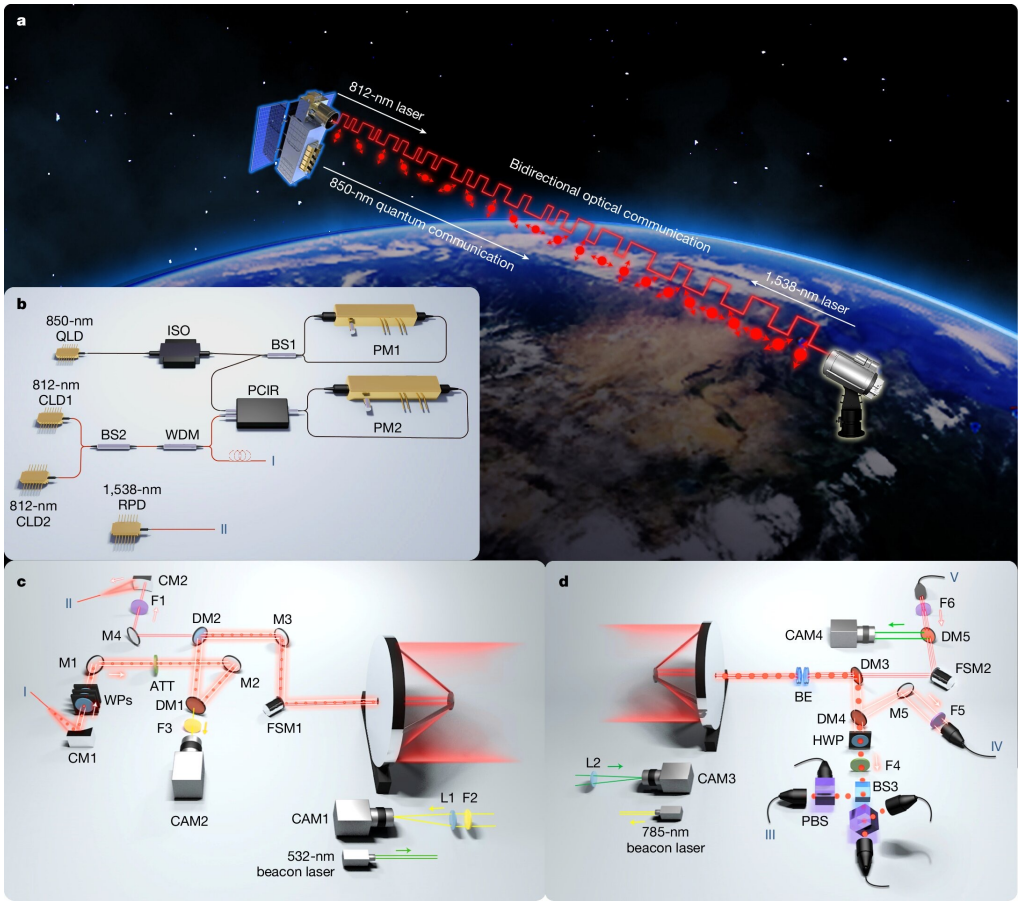
phys.org, USTC, University of Vienna, physicsworld.com

Quantum key distribution (QKD) ~ 2025

Intercontinental quantum communication via microsatellite and portable optical grounds stations, over 12,900 km, between China and South Africa.

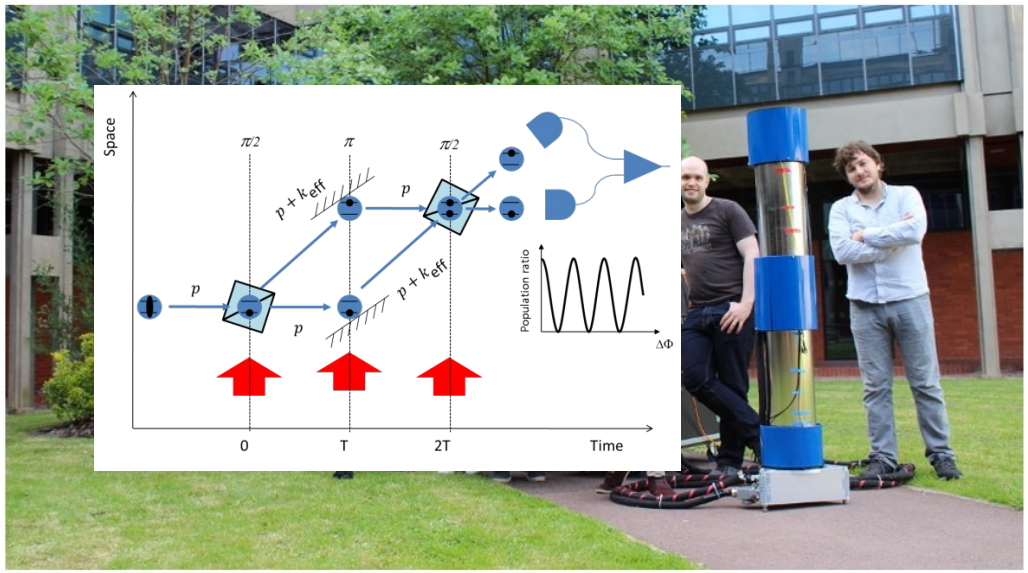
Portable optical ground stations are 10 x lighter compared with 2017 demonstration.

Microsatellite ~23 kg, compared with Micius ~250 kg.

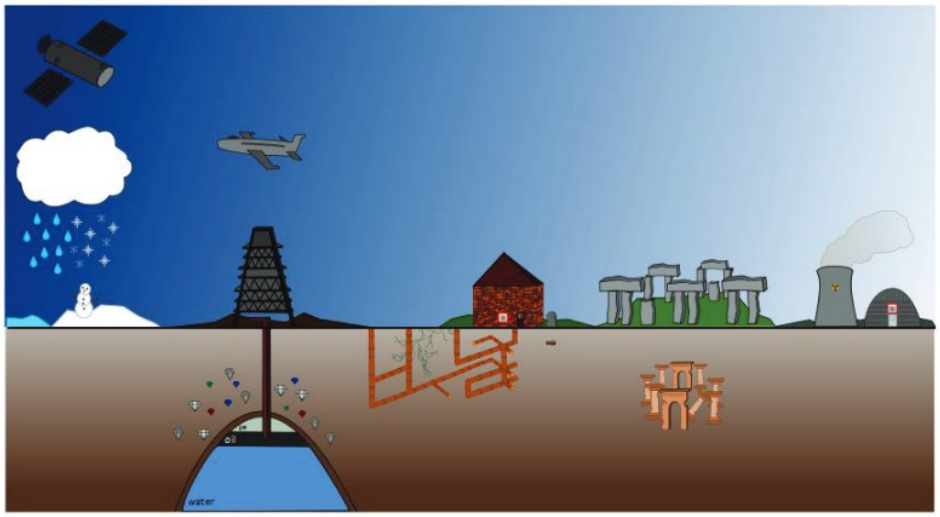


phys.org, Li et al. Nature 2025

Quantum Sensing



A gravity gradiometer with the Power and Infrastructure Research Group. Credit. University of Birmingham



Underground mapping, e.g. for

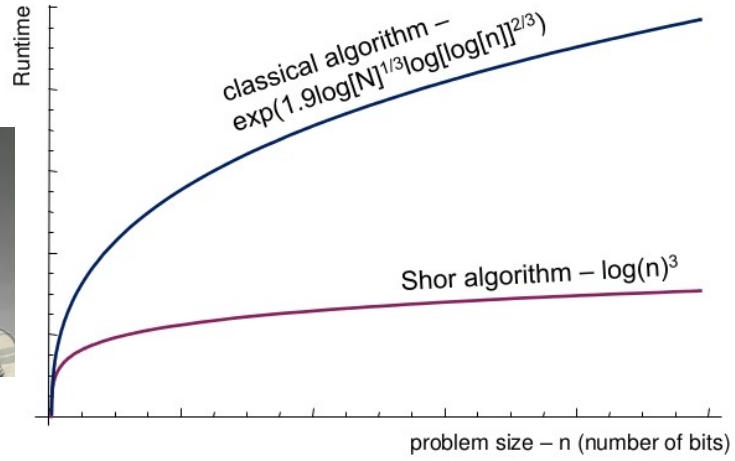
- Natural resource exploration
- Security

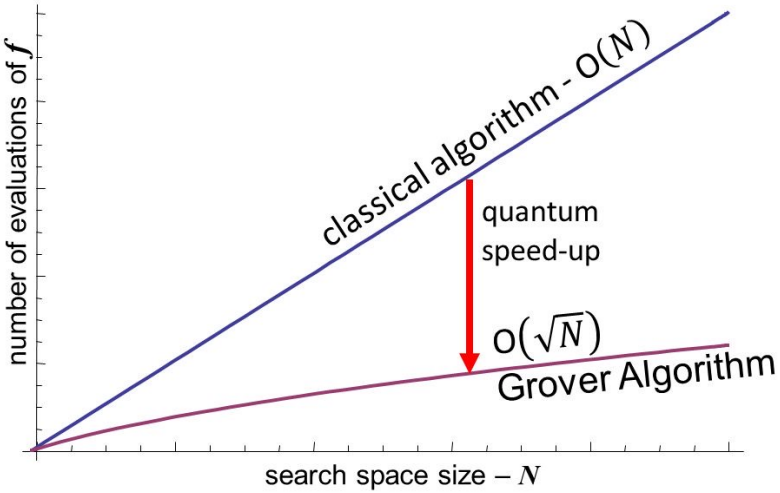
Quantum Computing

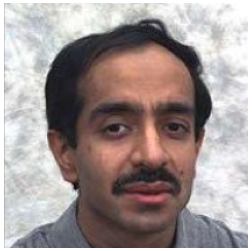
Quantum computers are **not** just faster versions of classical computers.

Quantum computers have known advantages *for very specific* problems.

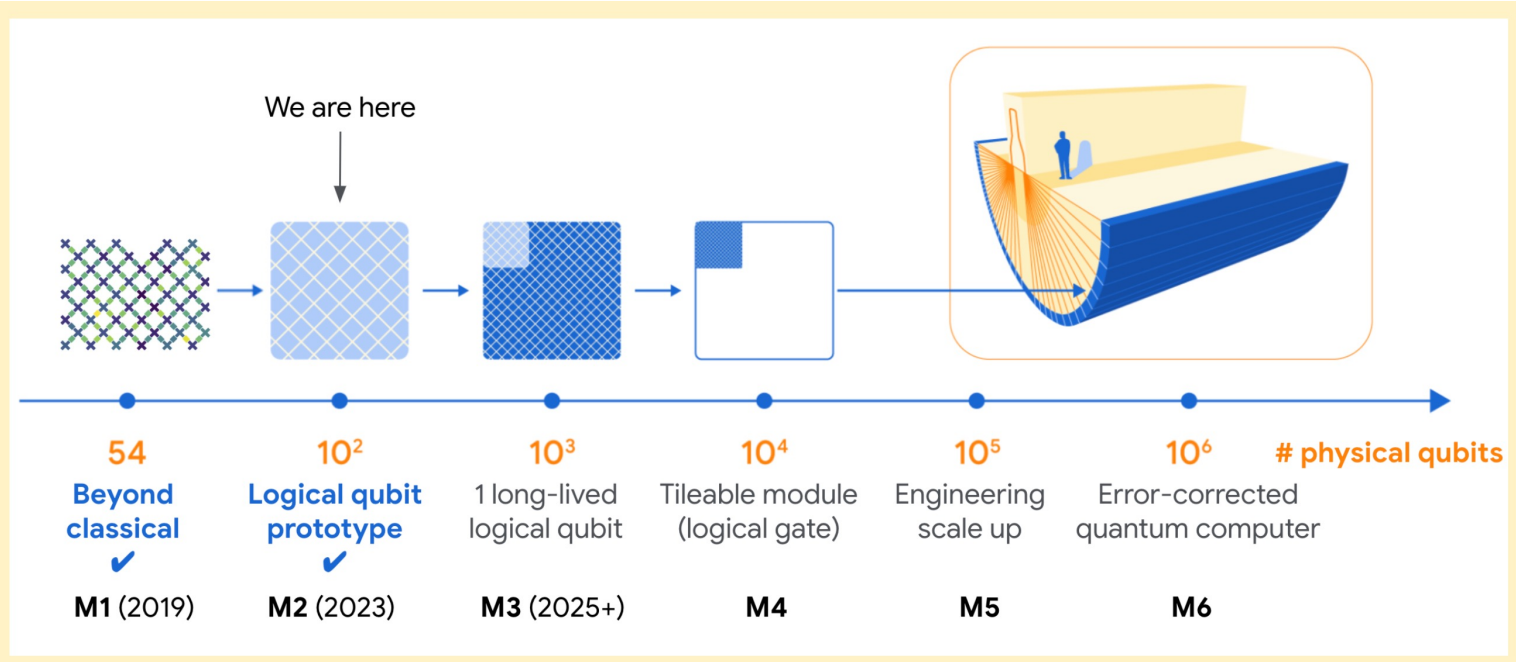






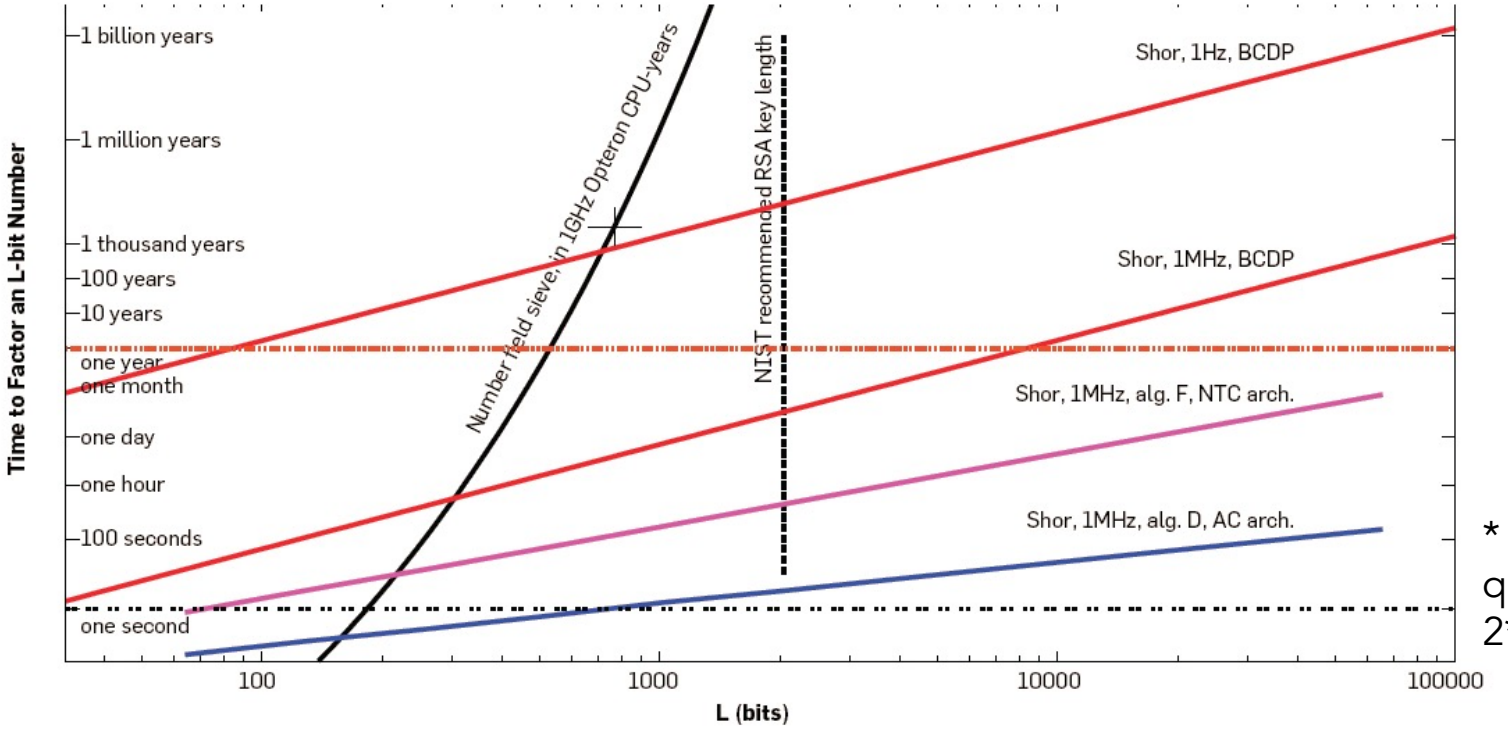


Current quantum computing



quantumai.google

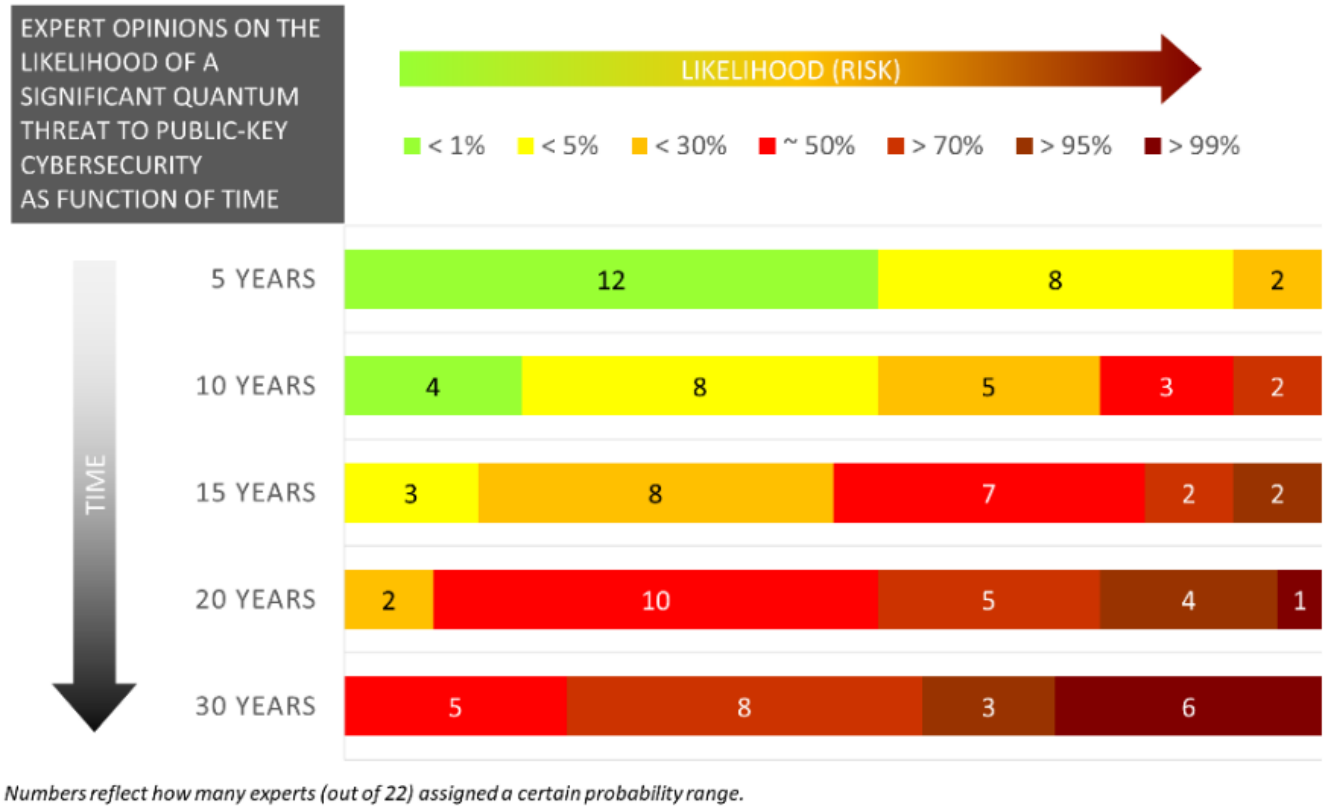
Why do we need millions of qubits?



* Shor estimates are for a quantum computer with $2*L^2$ ***logical*** qubits

Van Meter, Horsman, ACM 2013

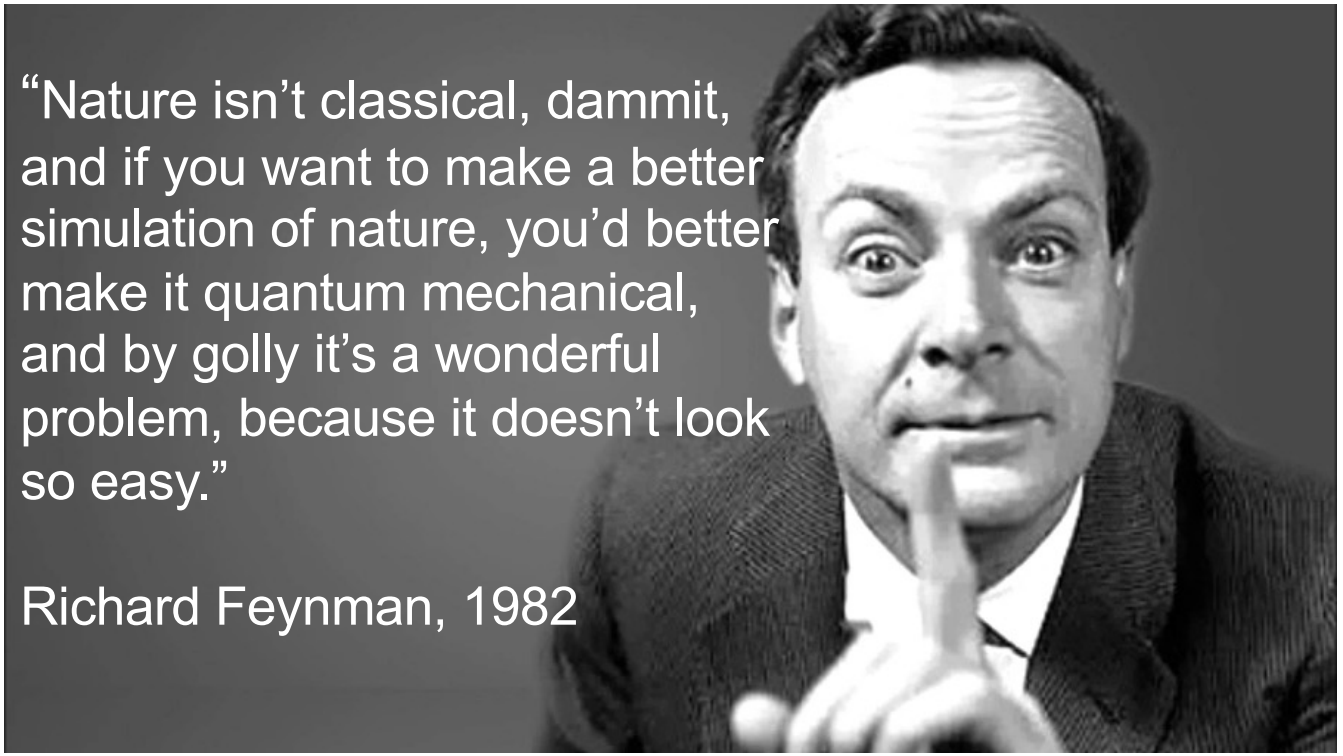
Quantum computers are a threat to our current cryptography



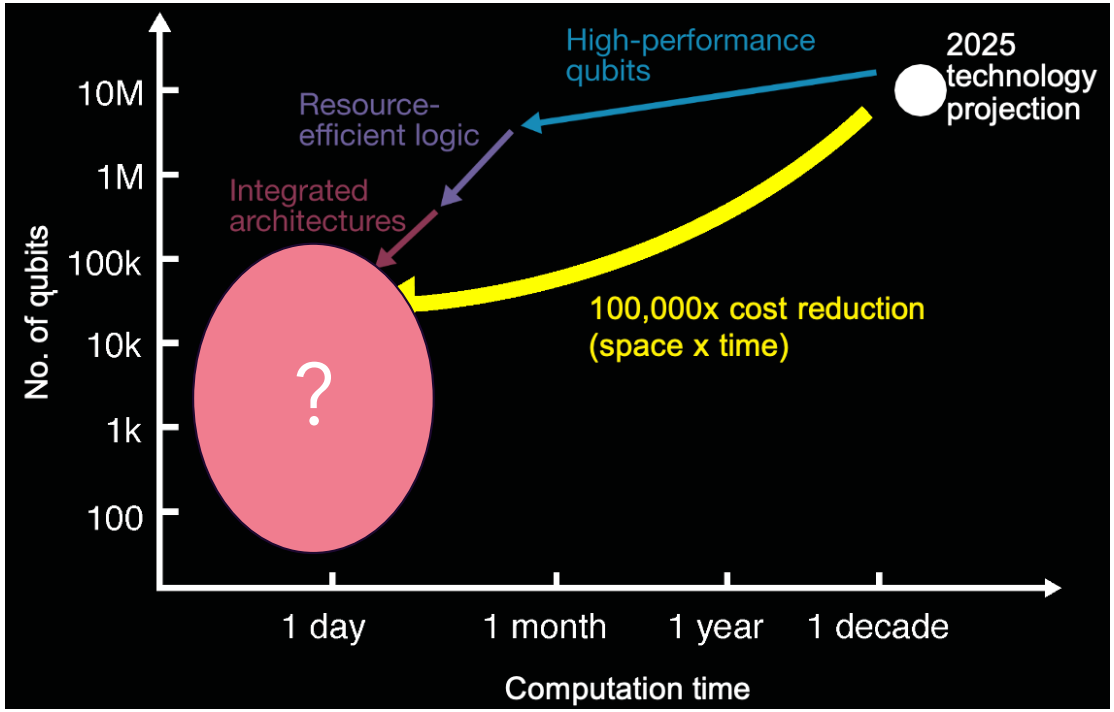
Numbers reflect how many experts (out of 22) assigned a certain probability range.

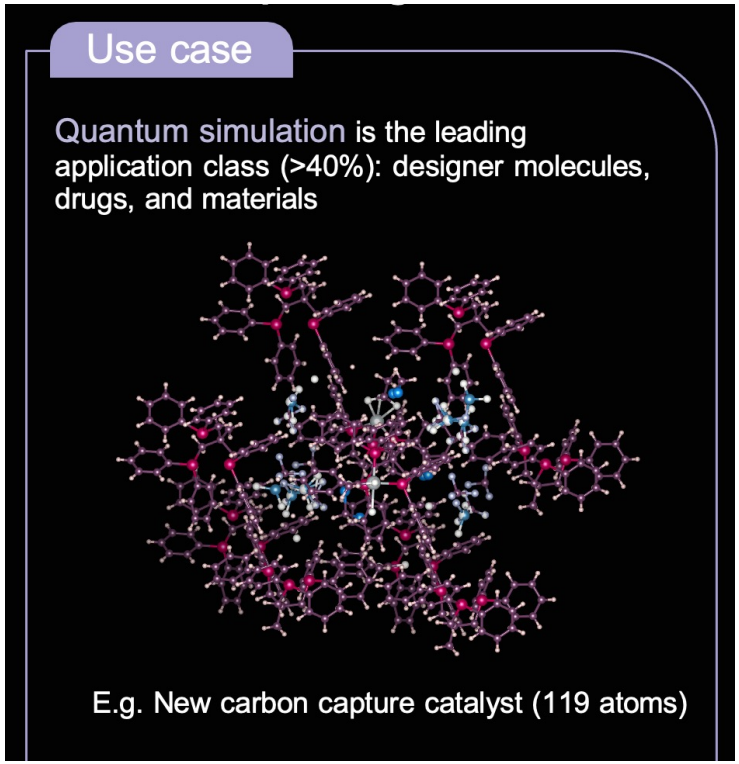
Mosca, Piani, Quantum Threat Timeline Report, 2019

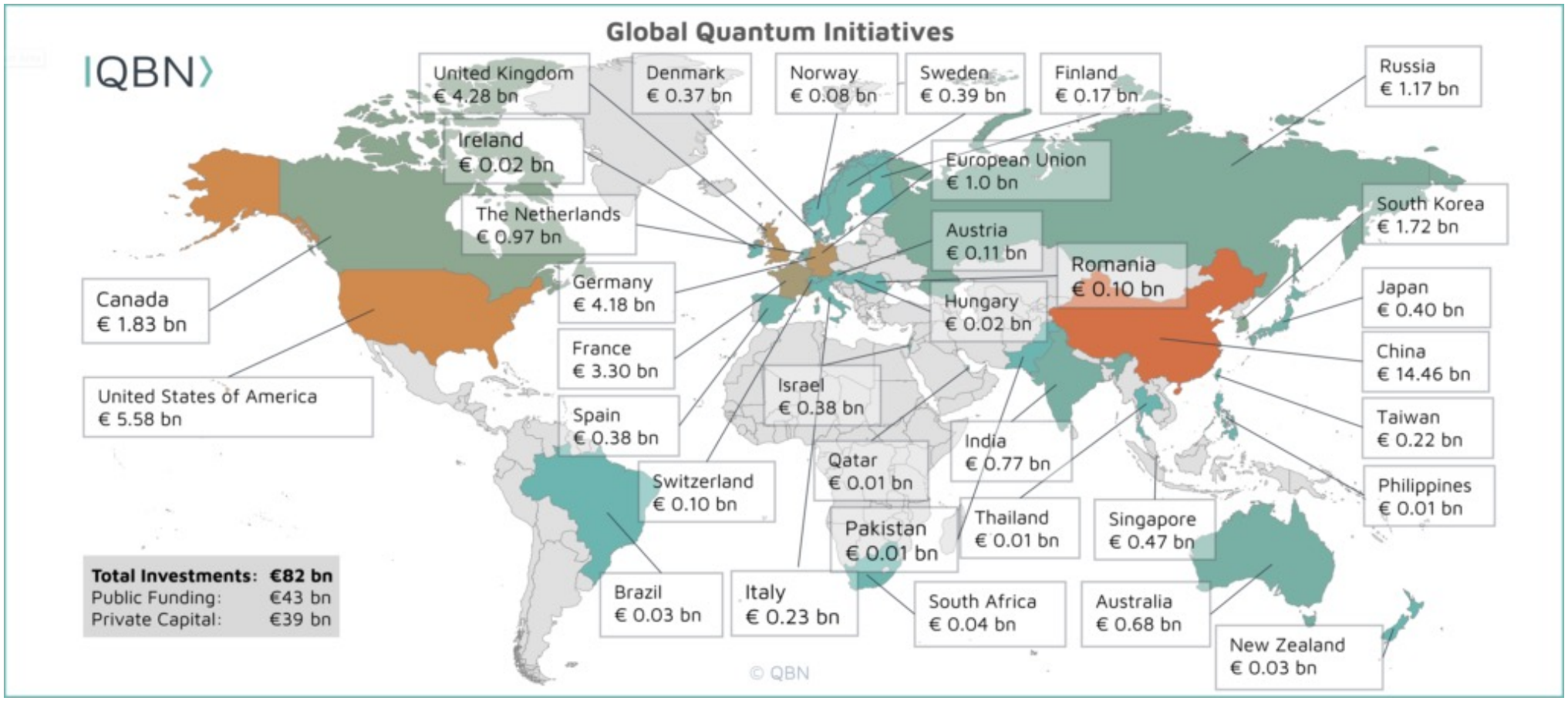
Nature is quantum…



We have quantum computers now, but they are not economically useful







https://qbn.world/intelligence/global-quantum-initiatives/

Einstein, Podolsky, Rosen 1935 – QM is incomplete

MAY 15, 1935 PHYSICAL REVIEW VOLUME 47

Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?

A. Einstein, B. Podolsky and N. Rosen, *Institute for Advanced Study, Princeton, New Jersey*

(Received March 25, 1935)

In a complete theory there is an element corresponding to each element of reality. A sufficient condition for the reality of a physical quantity is the possibility of predicting it with certainty, without disturbing the system. In quantum mechanics in the case of two physical quantities described by non-commuting operators, the knowledge of one precludes the knowledge of the other. Then either (1) the description of reality given by the wave function in quantum mechanics is not complete or (2) these two quantities cannot have simultaneous reality. Consideration of the problem of making predictions concerning a system on the basis of measurements made on another system that had previously interacted with it leads to the result that if (1) is false then (2) is also false. One is thus led to conclude that the description of reality as given by a wave function is not complete.



Ivan Kaminoff, flickr.com

While we have shown quantum mechanics does not provide a complete description of the physical reality, we left open the question of whether or not such a description exists. We believe however, that such a theory is possible.

Bell 1964: Can local hidden variables explain quantum correlations?

Realism: Measurement results are determined by "hidden variables" that exist prior to and independent of the experimental situation.

Locality: Results obtained at one location are independent of any measurements, or actions at another location.

Freedom of choice: Experimenters are free to choose which measurements to perform.



These assumptions are incompatible with quantum mechanics.

Physics Vol. 1, No. 3, pp. 195–200, 1964 Physics Publishing Co. Printed in the United States

ON THE EINSTEIN PODOLSKY ROSEN PARADOX*

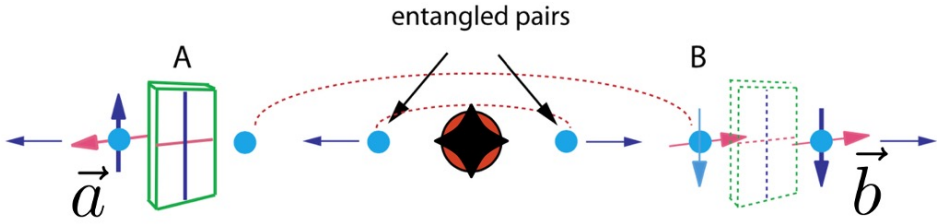
J. S. BELL†
Department of Physics, University of Wisconsin, Madison, Wisconsin

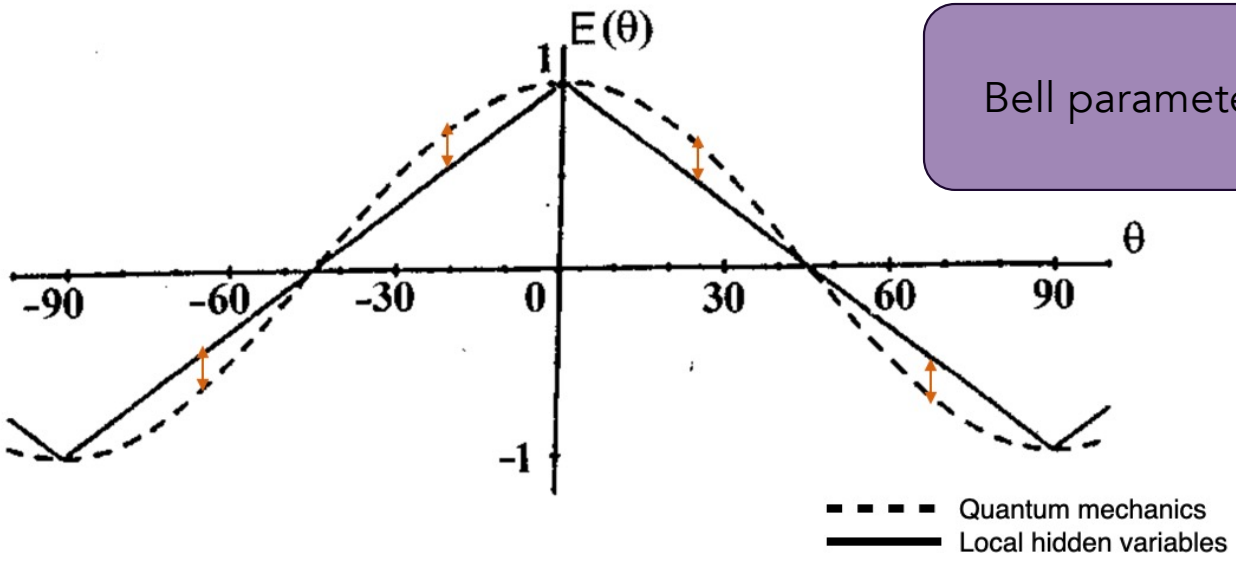
(*Received* 4 *November* 1964)

NO!

THE paradox of Einstein, Podolsky and Rosen was advanced as an argument that quantum mechanics could not be a complete theory but should be supplemented by additional variables. These additional variables were to restore to the theory causality and locality [2]. In this note that idea will be formulated mathematically and shown to be incompatible with the statistical predictions of quantum mechanics. It is the requirement of locality, or more precisely that the result of a measurement on one system be unaffected by operations on a distant system with which it has interacted in the past, that creates the essential difficulty. There have been attempts [3] to show that even without such a separability or locality requirement no "hidden variable" interpretation of quantum mechanics is possible. These attempts have been examined elsewhere [4] and found wanting. Moreover, a hidden variable interpretation of elementary quantum theory [5] has been explicitly constructed. That particular interpretation has indeed a grossly nonlocal structure. This is characteristic, according to the result to be proved here, of any such theory which reproduces exactly the quantum mechanical predictions.

Bell's inequality





Bell parameter: $|S_{LHV}| \leq 2$

$$|S_{QM}| = 2\sqrt{2}$$

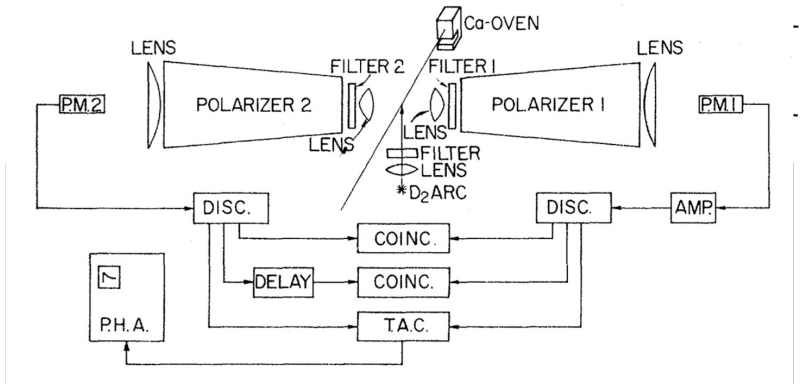
1972: Bell inequality violation in photons entangled in their polarisation

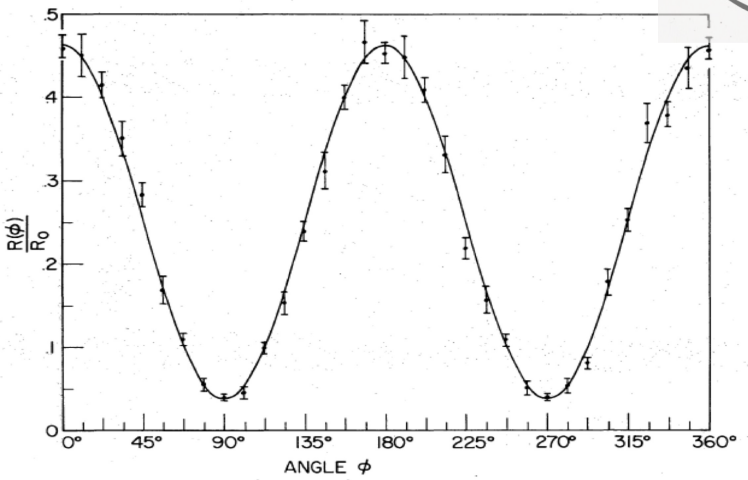


VOLUME 28, NUMBER 14 PHYSICAL REVIEW LETTERS 3 APRIL 1972

Experimental Test of Local Hidden-Variable Theories*

Stuart J. Freedman and John F. Clauser
Department of Physics and Lawrence Berkeley Laboratory, University of California, Berkeley, California 94720





Local hidden variable theories cannot explain quantum correlations.
The correlations measured agree with quantum mechanics.

Gisin's theorem: All entangled pure state of two qubits violate a Bell inequality.

Gisin, Phys. Lett. A, 154 (1991)